

the primary mass flow are unchanged for a given value of thrust, but the secondary mass flow increases. As a result, the scale factor (thrust divided by secondary mass flow) is reduced by an increase in exit area. Similarly, as the exit area decreases, the secondary mass flow decreases and the scale factor increases. Values as high as 115 lbf/lbm/sec (1128 N/kg/sec) can be obtained with the 80 percent exit area. Thus the scaling factor can be changed in two ways – by changing the value of thrust (varying the primary line pressure) or by changing the nozzle exit area.

Centerbody in the nozzle exit.– In most jet engines, there is a centerbody ahead of the nozzle exit, and consequently the exit is more annular than circular in cross section. The effect of the centerbody on the performance of the jet-engine simulator was investigated by testing the configuration sketched in figure 11. The results are compared with those for the basic ejector configuration in figure 12. Both nozzles have the same geometric exit area. It can be seen from these data that 15 percent additional primary-line total pressure and an additional increment of primary mass flow of about 0.14 lbm/sec (0.064 kg/sec) are required to achieve a particular value of thrust. The centerbody nozzle also has a reduced secondary mass flow. This increases the magnitude of the scale factor (thrust divided by secondary mass flow) by more than 40 percent, to values as high as 133 lbf/lbm/sec (1300 N/kg/sec).

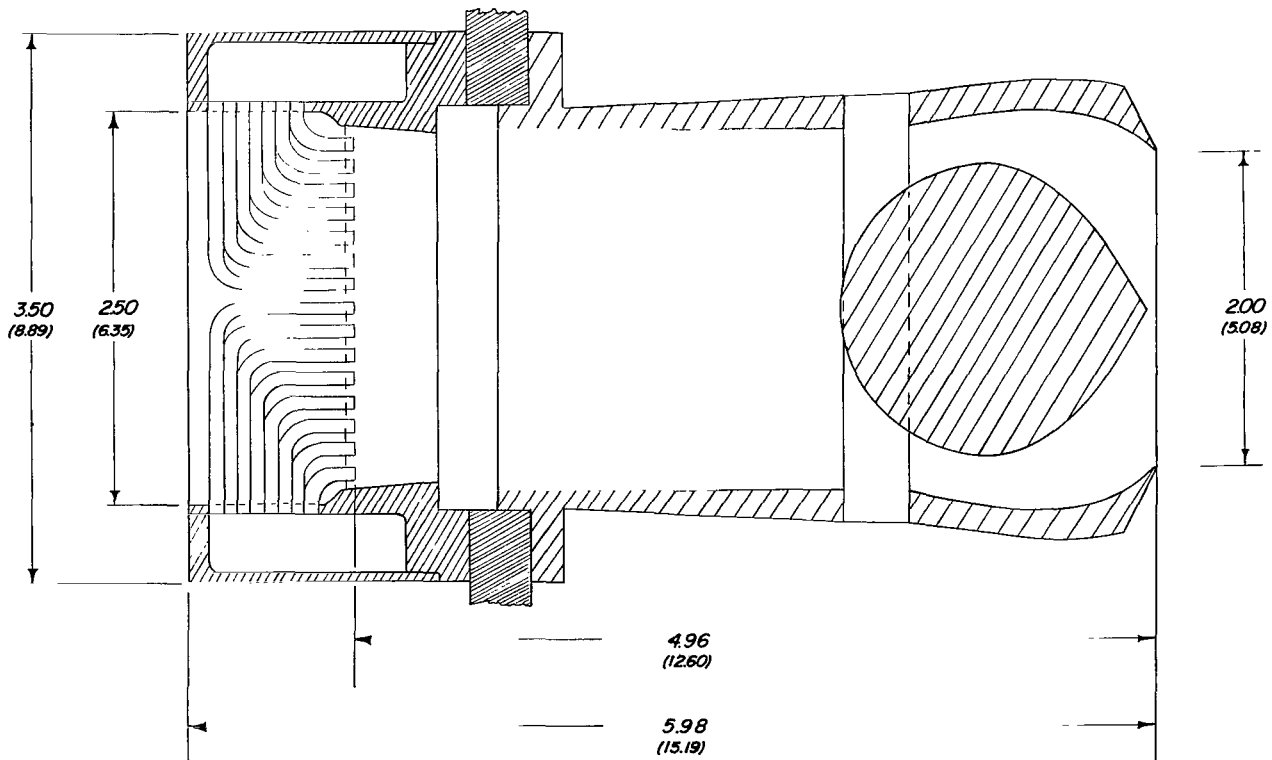


Figure 11.- Sketch of ejector with centerbody in the nozzle. Dimensions are in inches (centimeters).